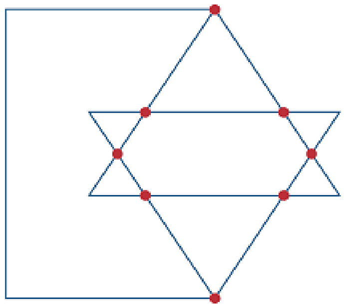
Chapter 2, Problem 6P

Problem

In the network graph shown in Fig. 2.70, determine the number of branches and nodes.

Figure 2.70:



Step-by-step solution

Step 1 of 6

Consider the following network graph:

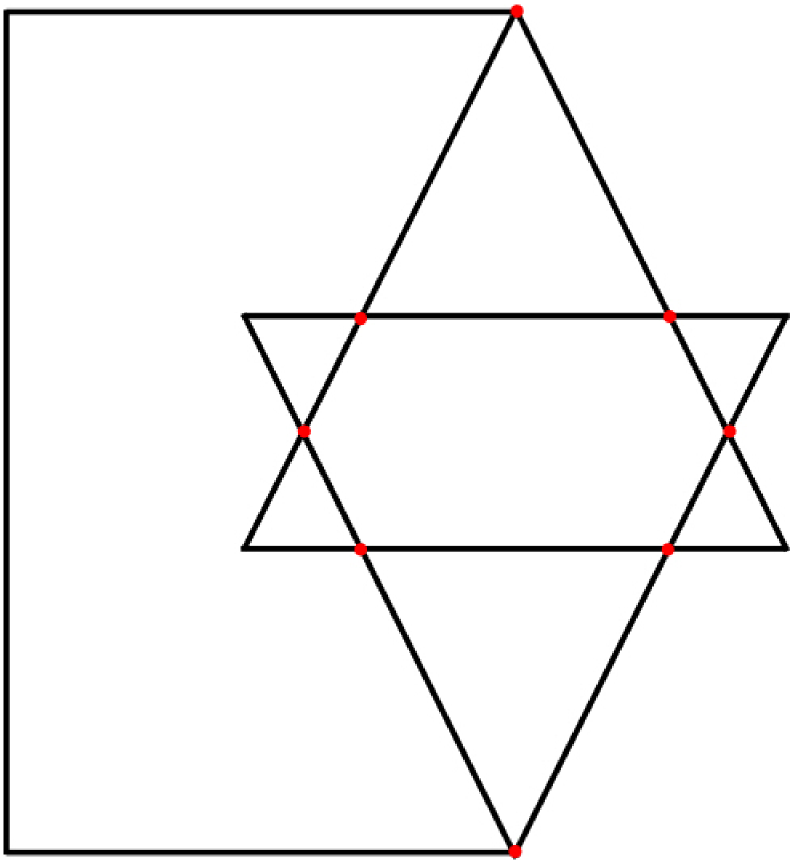


Figure 1

Step 2 of 6

Define a node:

A **node** is the point of connection between two or more branches.

Redraw the circuit with nodes.

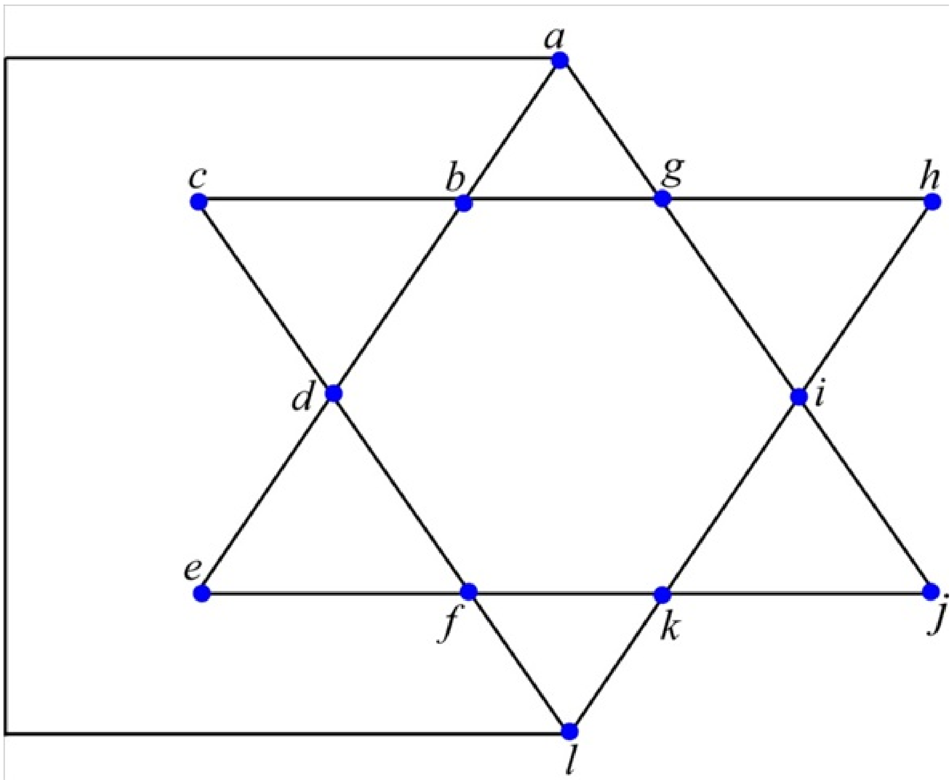


Figure 2

Step 3 of 6

We demonstrate that the circuit in Figure 1 has only **twelve** $(a \text{ to } l)$ nodes by redrawing the circuit in Figure 2.

Therefore number of nodes n in the circuit is $\boxed{12}$.

Step 4 of 6

Define a loop:

A **loop** is any closed path in a circuit.

Redraw the circuit with loops

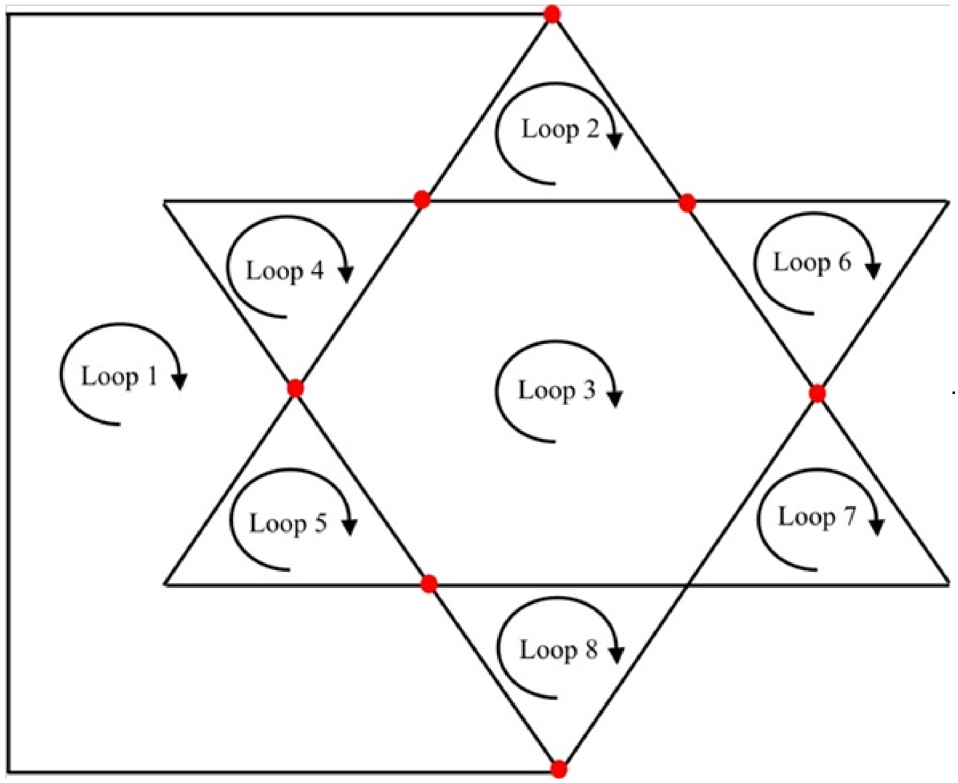


Figure 3

Step 5 of 6

We demonstrate that the circuit has only **eight** loops (loop 1 to loop 8) by redrawing the circuit in Figure 3.

Therefore number of loops l in the circuit is eight.

Step 6 of 6

Fundamental theorem of network topology states that if network with b branches, n nodes, and l independent loops will satisfy the following equation:

$$b = l + n - 1$$

A branch represents any two-terminal element.

Find the number of branches using fundamental theorem of network topology:

Substitute 8 for l and 12 for n in equation (1).

$$\begin{aligned} b &= 8 + 12 - 1 \\ &= 20 - 1 \\ &= 19 \end{aligned}$$

Therefore the number of branches b in the circuit is $\boxed{19}$.